
AYOT SafeSport™ Platform

AI-Powered Athlete Safety & Performance Infrastructure for Schools and Sports Clubs

Concept Note / Investment Proposal

Prepared by
Ayot Health Solutions

1. Executive Summary

AYOT SafeSport™ is an integrated **sports health, injury prevention, and athlete performance platform** designed for schools, sports academies, clubs, and tournament organizers.

The platform combines **AI-powered movement risk screening, sports medical coverage, digital athlete health records, and injury prevention programs** to provide structured medical oversight for youth sports.

Across Kenya, millions of students participate in sports each year, yet most institutions lack access to sports medicine services, athlete screening, and injury monitoring systems. This leads to preventable injuries, delayed recovery, and growing institutional liability risks.

AYOT SafeSport™ addresses this gap through a **scalable hybrid model combining digital technology and mobile sports medicine services**.

Key features include:

- AI-assisted movement and injury-risk screening
- Mobile sports medicine support during games and training
- Digital athlete health records and incident reporting
- Injury prevention and rehabilitation programs
- Institutional sports safety and compliance systems

The platform is designed to be **affordable, scalable, and deployable across Kenya's education and sports ecosystem**.

AYOT SafeSport™ aims to become the **national infrastructure platform for athlete health and safety in youth sports**.

2. Problem Statement

Sports participation among young people in Kenya is rapidly increasing due to school programs, youth academies, and national sports initiatives.

However, the sports ecosystem faces significant gaps:

Lack of Sports Medical Support

Most schools and clubs operate sports programs without professional medical oversight.

High Rates of Preventable Injuries

Non-contact injuries such as ankle sprains, ACL injuries, and muscle strains are common but often preventable.

No Athlete Health Monitoring

Schools rarely maintain structured records of athlete injuries, recovery progress, or return-to-play clearance.

Limited Access to Sports Science

Advanced injury risk screening and performance analysis are available only to elite athletes or professional teams.

Institutional Risk Exposure

Schools and tournament organizers face increasing legal and reputational risks when injuries occur without proper documentation or medical oversight.

These challenges highlight the urgent need for a **structured, scalable sports health system for youth athletes.**

3. Solution

AYOT SafeSport™ provides an integrated platform that enables schools and sports organizations to operate safer and more professional sports programs.

The platform integrates four core components.

3.1 AI Movement Risk Screening

Athletes perform standardized movement drills that are recorded using smartphone video.

AI-based pose estimation analyzes biomechanical movement patterns to identify injury risk factors such as:

- Knee valgus during landing
- Poor trunk stability
- Limb asymmetry
- Delayed movement stabilization

Each athlete receives a **personalized injury risk report** with corrective guidance.

Teams receive **risk dashboards** that help coaches identify injury patterns within their squads.

3.2 Sports Medical Coverage

AYOT SafeSport™ deploys trained clinicians to provide medical support during:

- Training sessions
- Competitive matches
- School sports days
- Tournaments

Services include:

- Injury assessment and first aid
 - Emergency response coordination
 - Medical referrals
 - Injury documentation and reporting
-

3.3 Athlete Digital Health Records

Each athlete receives a secure digital health profile containing:

- Screening results
- Injury history
- Rehabilitation progress

- Return-to-play clearance
- Performance testing results

This enables structured athlete health monitoring across seasons.

3.4 Injury Prevention & Performance Support

The platform supports athletes through:

- Pre-participation physical examinations
- Physiotherapy and rehabilitation
- Strength and conditioning guidance
- Athlete workload monitoring
- Nutrition and recovery education

This creates a **structured athlete health ecosystem for schools and clubs**.

4. Market Opportunity

Kenya presents a large and underserved market for structured sports health services.

Key indicators include:

- Over **90,000 schools nationwide**
- Approximately **17 million students**
- More than **1,000 sports clubs and academies**
- Rapid growth of youth sports competitions and tournaments

Despite this large ecosystem, structured sports medicine services are extremely limited.

AYOT SafeSport™ targets the following segments:

Schools

Primary and secondary schools with active sports programs.

Sports Academies

Football academies, athletics training centers, and other youth development programs.

Clubs

Community teams and league clubs.

Tournament Organizers

Inter-school competitions, county tournaments, and national youth events.

The total addressable market represents a significant opportunity for a **national sports health infrastructure platform**.

5. Business Model

AYOT SafeSport™ operates through a hybrid **subscription and service-based model**.

Revenue streams include:

Institutional Subscriptions

Schools and academies subscribe to SafeSport plans that provide screening, medical coverage, and athlete monitoring.

Event Medical Coverage

Medical teams provide coverage for sports tournaments and competitions.

Athlete Screening Programs

Teams pay for AI movement risk screening sessions.

Physiotherapy and Rehabilitation

Clinics provide injury management and recovery services.

Digital Platform Licensing

Institutions may license access to the athlete health record and risk monitoring system.

6. Product Plans

The SafeSport platform offers three scalable service tiers.

SafeSport Basic

Entry-level coverage for smaller schools and clubs.

Includes:

- On-field first aid support
 - Injury documentation
 - Basic musculoskeletal screening
 - Injury prevention education
-

SafeSport Performance

Designed for competitive schools and academies.

Includes:

- Pre-participation physical exams
 - Physiotherapy clinics
 - Injury management plans
 - Athlete readiness reports
-

SafeSport Elite

High-performance support for elite institutions.

Includes:

- Weekly rehabilitation clinics
 - Athletic performance profiling
 - Return-to-play protocols
 - Mental performance coaching
 - Performance analytics dashboards
-

7. Technology Platform

The SafeSport digital infrastructure supports all service delivery.

Key features include:

Field Clinician Mobile Application

Clinicians record injuries, assessments, and treatment notes directly from the field.

Athlete Health Record System

Athlete profiles track injury history, recovery progress, and readiness status.

AI Movement Screening Engine

Video-based biomechanical analysis generates athlete risk scores.

Client Portal

Schools and sports administrators access dashboards showing injury trends and athlete status.

Reporting and Compliance Tools

Automated injury reports and incident documentation support institutional safety and compliance.

The platform integrates with the broader **AYOT digital health infrastructure**.

8. Competitive Advantage

AYOT SafeSport™ offers several unique advantages.

First-Mover Advantage

Few organizations provide structured sports medicine systems for schools in Kenya.

Integrated Digital Platform

The platform combines injury prevention, medical services, and athlete data systems.

Scalable Service Model

Mobile clinician networks enable nationwide deployment.

Data-Driven Approach

Athlete health and injury data provide valuable insights for prevention and performance.

Affordable Pricing Structure

Tiered plans make sports medicine accessible to institutions of different sizes.

9. Implementation Strategy

The rollout strategy includes three phases.

Phase 1 – Pilot Programs

Deploy pilots with selected schools and academies to validate service delivery.

Phase 2 – Regional Expansion

Establish regional clinician networks in key sports hubs.

Phase 3 – National Platform Scaling

Expand services across schools, academies, and sports federations.

10. Impact

AYOT SafeSport™ is designed to generate measurable impact.

Expected outcomes include:

- Reduction in preventable sports injuries
 - Improved athlete recovery and safer return to play
 - Increased access to sports medicine services
 - Safer sports participation for young athletes
 - Strengthened sports programs within schools and communities
-

11. Investment Opportunity

AYOT SafeSport™ seeks investment to support:

- Technology platform development
- AI screening system enhancement
- Pilot deployments across schools and academies
- Recruitment and training of sports medicine clinicians
- Market expansion and partnerships

The investment will enable AYOT SafeSport™ to scale as **Kenya's leading sports health infrastructure platform.**

12. Vision

AYOT SafeSport™ aims to ensure that every young athlete has access to safe sports participation and professional health support.

Our long-term goal is to establish a **national athlete safety and performance ecosystem** that empowers institutions, protects athletes, and advances the future of sports health.

AYOT SafeSport Platform

Technical Specification Document (TSD)

Version 1.0 — Implementation Ready

Prepared for:

Ayot Health Solutions

System Name:

AYOT SafeSport Platform

Purpose:

To deliver an integrated **AI-assisted athlete injury prevention, sports medicine service management, and athlete health record platform** for schools, sports academies, and clubs.

1. System Overview

1.1 Product Definition

The **AYOT SafeSport Platform** is a modular digital system that enables sports institutions to manage athlete health, injury prevention, and sports medical services through an integrated platform combining:

1. AI-assisted movement risk screening
2. Athlete electronic medical records (EMR)
3. Sports medical service scheduling
4. Incident documentation and reporting
5. Performance monitoring
6. Institutional sports safety dashboards

The platform integrates mobile clinician tools, athlete screening analytics, and institutional management dashboards.

2. System Architecture

2.1 Architecture Model

The system follows a **modular cloud-based architecture**.

Architecture Layers:

1. **User Interface Layer**
 2. **Application Services Layer**
 3. **AI Analytics Engine**
 4. **Data Storage Layer**
 5. **Integration & API Layer**
-

2.2 High-Level Architecture Diagram

User Layer

- Athlete Portal (optional)
- Coach / School Dashboard
- Clinician Mobile App
- Admin Control Panel

Application Layer

- Client Management Service
- Athlete EMR Service
- Screening Management Service
- Incident Reporting Service
- Scheduling & Rostering
- Billing & Subscription

AI Analytics Layer

- Pose Estimation Engine
- Biomechanics Risk Scoring Engine
- Performance Analytics Engine

Data Layer

- Athlete Database
- Incident Database
- Screening Dataset Storage
- Video Storage

Integration Layer

- AYOT Backbone API
 - Insurance Systems
 - Laboratory Systems
 - Telemedicine Services
-

3. Core System Modules

3.1 Client & Institution Management Module

Manages institutions subscribing to the SafeSport platform.

Functions

- Register schools/clubs
- Manage institution contacts
- Manage subscription plans
- Track active teams and sports

Data Model

clients

field	type
client_id	UUID
name	string
type	enum (school, club, academy)
contact_person	string
phone	string
email	string
subscription_plan	enum
status	active/inactive

3.2 Athlete Electronic Medical Record (EMR)

Stores health and performance data for athletes.

Core Features

- Athlete profiles
- Injury history
- Screening results
- Rehabilitation tracking
- Return-to-play status

Athlete Table

field	type
athlete_id	UUID
name	string
date_of_birth	date
gender	string
sport	string
team	string
guardian_contact	string
medical_notes	text
allergies	text

3.3 Screening Management System

Manages AI-assisted movement screening sessions.

Screening Workflow

1. Athlete registered
 2. Screening session created
 3. Movement drills recorded
 4. Video uploaded
 5. AI pose estimation processed
 6. Risk scores calculated
 7. Physiotherapist review
 8. Report generated
-

3.4 Movement Screening Protocol

Standard drills include:

- Jump landing test
- Single leg squat
- Sprint acceleration
- Cutting maneuver
- Kicking mechanics

Video capture requirements:

Resolution: **minimum 1080p**

Frame rate: **minimum 30fps**

Camera distance: **3–5 meters**

Lighting: **natural or well-lit environment**

4. AI Analytics Engine

4.1 Pose Estimation

Recommended model:

Google MoveNet

Alternative:

MediaPipe Pose

Output:

17–33 body keypoints per frame.

Example:

```
{  
  "nose": (x,y),  
  "left_shoulder": (x,y),  
  "right_shoulder": (x,y),  
  "left_knee": (x,y),  
  "right_knee": (x,y)  
}
```

4.2 Biomechanical Metrics

Metrics calculated:

Knee Valgus Angle

Detects knee inward collapse during landing.

Thresholds

Low Risk: $<5^{\circ}$
Moderate Risk: $5\text{--}10^{\circ}$
High Risk: $>10^{\circ}$

Trunk Lean Angle

Measures torso stability.

Low Risk: $<10^{\circ}$
Moderate Risk: $10\text{--}20^{\circ}$
High Risk: $>20^{\circ}$

Limb Symmetry Index

Measures bilateral performance.

$LSI = \text{weaker limb} / \text{stronger limb} \times 100$

Low Risk: $\geq 90\%$
Moderate Risk: $80\text{--}89\%$
High Risk: $<80\%$

Time to Stabilization

Measures landing control.

Low Risk: <0.5 seconds
Moderate Risk: $0.5\text{--}1.0$ seconds
High Risk: >1.0 seconds

4.3 Composite Risk Score

Each metric assigned:

Green = 0
Amber = 1
Red = 2

Total Risk Score:

Score Classification

0–2 Low Risk
3–5 Moderate Risk
6+ High Risk

5. Incident Reporting System

Used during sports events or clinics.

Incident Data

field	type
incident_id	UUID
athlete_id	UUID
clinician_id	UUID
location	string
sport	string
injury_type	string
severity	enum
treatment	text
photo_url	string
timestamp	datetime

Severity Levels

- Minor
 - Moderate
 - Severe
 - Emergency
-

6. Scheduling & Operations Module

Manages clinician deployment.

Functions:

- Event bookings
- Clinic scheduling
- Staff assignment
- Equipment checklist

Schedules Table

field	type
schedule_id	UUID
client_id	UUID
site	string
clinician_id	UUID
start_time	datetime
end_time	datetime
service_type	event/clinic/screening

7. Billing & Subscription System

Supports:

- recurring subscriptions
- event charges
- add-on services

Billing Table

field	type
invoice_id	UUID
client_id	UUID
amount	decimal
service_type	string
status	unpaid/paid
date_issued	datetime

8. Mobile Clinician App

Primary interface for field clinicians.

Key Screens:

1. Login
2. Today's Schedule
3. Athlete Lookup
4. Screening Data Entry
5. Injury Incident Form
6. Referral Request
7. Sync Status

Features:

- Offline data capture
 - Photo attachments
 - e-signatures
 - automatic syncing
-

9. Client Dashboard (Web)

Accessible to schools and administrators.

Dashboard features:

- injury trends
 - athlete readiness status
 - screening results
 - upcoming medical visits
 - incident reports
 - invoice history
-

10. Security & Compliance

Compliance requirements:

Kenyan Data Protection Act (2019)

Requirements:

- parental consent for minors
- secure cloud storage

- data access controls
-

Security Measures

Encryption:

TLS for transmission

AES-256 for storage

Authentication:

JWT token authentication

Access control:

Role-based permissions

11. System Infrastructure

Recommended deployment stack:

Frontend

React / Next.js

Mobile

Flutter or React Native

Backend

Python (FastAPI) or Node.js

Database

PostgreSQL

AI Processing

Python

Cloud Platform

Google Cloud or AWS

Video Storage

Cloud object storage (S3 / Google Cloud Storage)

12. Implementation Timeline

Phase 1 — System Design

Duration: 3 weeks

Deliverables:

- architecture finalization
 - database schema
 - UI wireframes
-

Phase 2 — MVP Development

Duration: 10–12 weeks

Build:

- clinician mobile app
 - screening engine
 - incident reporting
 - client dashboard
-

Phase 3 — Pilot Deployment

Duration: 6 weeks

Deploy with:

3–5 schools or clubs

Objectives:

- validate workflows
 - collect injury data
 - refine UI
-

Phase 4 — Platform Expansion

Duration: ongoing

Add:

- telemedicine
 - insurance integrations
 - advanced analytics
-

13. Key Performance Indicators

Monitor platform performance using:

Operational

- number of athletes screened
- number of incidents documented
- clinician response time

Clinical

- injury incidence rate
- return-to-play duration

Business

- number of institutional subscriptions
 - revenue per client
 - subscription renewal rate
-

14. Minimum Deployment Requirements

Pilot deployment requires:

Personnel

- 1 sports physician
- 2 physiotherapists
- 1 operations coordinator
- 1 software engineer

Infrastructure

- clinician tablets
 - screening cameras
 - mobile network connectivity
-

15. Deliverables for Development Team

Engineering deliverables:

- API specification
- database schema
- clinician mobile app
- AI screening pipeline
- client dashboard

Operational deliverables:

- clinician training manual
 - screening protocol guide
 - data protection framework
-

1. AYOT SafeSport Platform

API Specification (v1.0)

API Style

RESTful JSON API

Base URL

`https://api.ayotsafesport.com/v1`

Authentication

JWT Token Authentication

Header format:

`Authorization: Bearer <token>`

1.1 Authentication Endpoints

Login

POST `/auth/login`

Request:

```
{
  "email": "clinician@ayot.com",
  "password": "password123"
}
```

Response:

```
{
  "token": "jwt_token",
  "user_id": "user_23",
  "role": "clinician"
}
```

1.2 Client Management

Create Client

POST /clients

Request:

```
{
  "name": "Green Valley School",
  "type": "school",
  "contact_person": "John Doe",
  "phone": "+254712000000",
  "email": "admin@greenvalley.ac.ke",
  "subscription_plan": "standard"
}
```

List Clients

GET /clients

Response:

```
[
  {
    "client_id": "cl_001",
    "name": "Green Valley School",
    "type": "school",
    "subscription_plan": "standard",
    "status": "active"
  }
]
```

1.3 Athlete Management

Create Athlete

POST /athletes

Request:

```
{
```

```
"name": "David Kimani",  
"date_of_birth": "2008-04-15",  
"gender": "male",  
"sport": "football",  
"team": "U16",  
"guardian_contact": "+254700000000"  
}
```

Get Athlete Profile

GET /athletes/{athlete_id}

Response:

```
{  
  "athlete_id": "ath_101",  
  "name": "David Kimani",  
  "sport": "football",  
  "team": "U16",  
  "injury_history": [],  
  "screenings": []  
}
```

1.4 Screening Module

Create Screening Session

POST /screenings

```
{  
  "team": "U16",  
  "sport": "football",  
  "date": "2026-05-10",  
  "location": "Green Valley School"  
}
```

Upload Screening Video

POST /screenings/{id}/video

Multipart upload

Fields:

file: video.mp4

```
athlete_id: ath_101
drill_type: jump_landing
```

Get Screening Results

GET /screenings/{id}/results

Response:

```
{
  "athlete_id": "ath_101",
  "knee_valgus_angle": 12.5,
  "trunk_lean": 15,
  "limb_symmetry_index": 82,
  "risk_score": 5,
  "risk_level": "moderate"
}
```

1.5 Incident Reporting

Create Incident

POST /incidents

```
{
  "athlete_id": "ath_101",
  "clinician_id": "user_23",
  "sport": "football",
  "injury_type": "ankle sprain",
  "severity": "moderate",
  "treatment": "compression bandage applied"
}
```

Get Incident Report

GET /incidents/{id}

1.6 Scheduling

Create Event Booking

POST /events

```
{
  "client_id": "cl_001",
  "event_name": "Inter-school football match",
  "date": "2026-06-01",
  "location": "Green Valley School",
  "service_type": "event_coverage"
}
```

1.7 Billing

Generate Invoice

POST /billing/invoice

```
{
  "client_id": "cl_001",
  "service_type": "event_coverage",
  "amount": 15000
}
```

2. Database Schema (SQL Ready)

Below is a PostgreSQL schema.

Clients Table

```
CREATE TABLE clients (
  client_id UUID PRIMARY KEY,
  name VARCHAR(255),
  type VARCHAR(50),
  contact_person VARCHAR(255),
  phone VARCHAR(50),
  email VARCHAR(255),
  subscription_plan VARCHAR(50),
  status VARCHAR(50),
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

Users Table

```
CREATE TABLE users (  
    user_id UUID PRIMARY KEY,  
    name VARCHAR(255),  
    email VARCHAR(255),  
    password_hash TEXT,  
    role VARCHAR(50),  
    phone VARCHAR(50),  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

Athletes Table

```
CREATE TABLE athletes (  
    athlete_id UUID PRIMARY KEY,  
    name VARCHAR(255),  
    date_of_birth DATE,  
    gender VARCHAR(20),  
    sport VARCHAR(50),  
    team VARCHAR(50),  
    guardian_contact VARCHAR(50),  
    medical_notes TEXT,  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

Screening Sessions

```
CREATE TABLE screenings (  
    screening_id UUID PRIMARY KEY,  
    client_id UUID,  
    sport VARCHAR(50),  
    team VARCHAR(50),  
    date DATE,  
    location VARCHAR(255),  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

Screening Results

```
CREATE TABLE screening_results (  
    result_id UUID PRIMARY KEY,  
    screening_id UUID,  
    athlete_id UUID,  
    knee_valgus FLOAT,  
    trunk_lean FLOAT,  
    limb_symmetry FLOAT,  
    stabilization_time FLOAT,  
    risk_score INT,
```

```
        risk_level VARCHAR(20)
    );
```

Incidents

```
CREATE TABLE incidents (
    incident_id UUID PRIMARY KEY,
    athlete_id UUID,
    clinician_id UUID,
    injury_type VARCHAR(100),
    severity VARCHAR(50),
    treatment TEXT,
    location VARCHAR(255),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

Events

```
CREATE TABLE events (
    event_id UUID PRIMARY KEY,
    client_id UUID,
    event_name VARCHAR(255),
    date DATE,
    location VARCHAR(255),
    service_type VARCHAR(50),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

Billing

```
CREATE TABLE invoices (
    invoice_id UUID PRIMARY KEY,
    client_id UUID,
    amount DECIMAL,
    service_type VARCHAR(50),
    status VARCHAR(50),
    issued_date DATE
);
```

3. Clinician Mobile App Wireframes

Below are the core screens for the field clinician app.

Screen 1 — Login

Fields:

Email

Password

Buttons:

Login

Screen 2 — Today's Schedule

TODAY'S ASSIGNMENTS

1. Green Valley School
Football Match
09:00 - 11:00

2. Nakuru Academy
Screening Session
14:00 - 16:00

Buttons:

Start Event

Screen 3 — Athlete Lookup

Search bar

Search athlete name

Results:

David Kimani - U16
John Otieno - U18

Button:

[View Profile](#)

Screen 4 — New Injury Incident

Fields:

Athlete Name

Sport

Injury Type

Severity

Severity slider:

Minor

Moderate

Severe

Treatment notes

Photo upload

Signature capture

Button:

Submit Report

Screen 5 — Screening Capture

Athlete:

Dropdown

Drill:

Jump Landing

Sprint

Cutting

Single Leg Squat

Buttons:

Record Video

Upload Video

Screen 6 — Sync Dashboard

Pending Uploads: 4

Last Sync: 10:45 AM

Buttons:

Sync Now

4. AI Screening Pipeline (Engineering Flow)

Video Upload



Frame Extraction



Pose Estimation (MoveNet)



Keypoint Tracking



Biomechanical Calculations



Risk Score Computation



Physiotherapist Validation



PDF Athlete Report



Team Risk Dashboard

5. MVP Development Roadmap

Month 1–2

Build:

- database
- authentication

- athlete EMR
 - incident reporting
-

Month 3

Build:

- screening pipeline
 - clinician mobile app
 - dashboards
-

Month 4

Pilot with:

3–5 schools

What This Enables

With this system in place, Ayot can operate:

- sports medicine services
- athlete screening programs
- injury data analytics
- institutional safety reporting

on one unified digital platform.

AYOT SafeSport AI Engine

Injury Prediction & Athlete Risk Intelligence Architecture

Version 1.0
Technical AI System Specification

1. System Objective

The **AYOT Injury Prediction Engine (IPE)** predicts the likelihood of sports injuries using biomechanical movement patterns, athlete characteristics, and training load data.

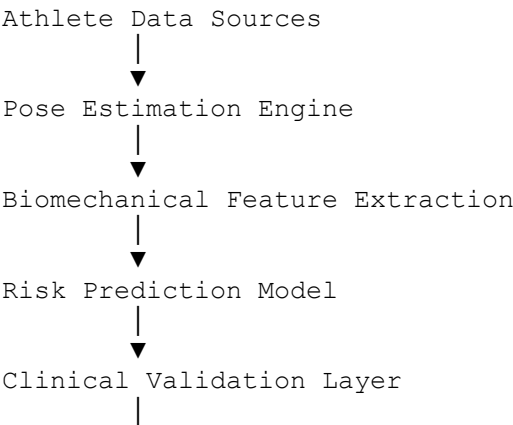
The system provides:

- Athlete injury risk probability
- Identification of biomechanical risk factors
- Early warning alerts for coaches and clinicians
- Injury prevention recommendations

This transforms SafeSport from **reactive injury management** to **predictive injury prevention**.

2. AI Engine Architecture

The system consists of **five machine-learning components**.



3. Data Inputs

The AI engine combines **three major datasets**.

3.1 Biomechanical Movement Data

Derived from screening videos.

Features extracted:

Feature	Description
knee_valgus_angle	inward knee collapse
trunk_lean	torso stability
hip_knee_ratio	hip alignment
landing_force_estimate	landing load
limb_symmetry_index	strength imbalance
stabilization_time	neuromuscular control

These features are extracted from **pose estimation keypoints**.

3.2 Athlete Profile Data

Variable	Description
age	growth stage risk
gender	injury pattern differences
sport	sport-specific risk
playing_position	positional injury risk
previous_injuries	recurrence risk
training_hours	fatigue exposure

3.3 Training Load Data (Optional)

Metric	Description
weekly_training_load	training intensity
game_minutes	match exposure
recovery_days	recovery time
fatigue_score	self-reported fatigue

4. Feature Engineering

Raw pose estimation coordinates are transformed into **biomechanical metrics**.

Example:

Knee Valgus Angle

Computed using joint coordinates.

$$\theta = \cos^{-1}\left(\frac{(\text{hip-knee}) \cdot (\text{ankle-knee})}{|\text{hip-knee}| |\text{ankle-knee}|}\right)$$

This measures the angle between hip-knee and ankle-knee vectors.

Limb Symmetry Index

$$LSI = \frac{\text{weaker limb}}{\text{stronger limb}} \times 100$$

Used to detect strength imbalance.

Stabilization Time

Measured from landing frame until center-of-mass oscillation stabilizes.

5. Injury Prediction Model

The prediction system uses **ensemble machine learning models**.

Recommended architecture:

Stage 1 — Feature Classification Model

Model type:

Gradient Boosted Trees (XGBoost)

Purpose:

Classify biomechanical risk features.

Output:

`biomechanical_risk_score`

Stage 2 — Injury Probability Model

Model type:

Random Forest

Input:

- biomechanical features
- athlete profile
- training load

Output:

`probability_of_injury_next_3_months`

Example output:

Risk Probability: 0.32
Risk Category: Moderate

Stage 3 — Deep Learning Movement Model

Used for long-term system improvement.

Model type:

Temporal Convolutional Neural Network

Input:

pose estimation sequences

Purpose:

detect subtle movement abnormalities.

6. Risk Classification System

Risk categories:

Risk Score Classification

0–0.2	Low Risk
0.21–0.4	Moderate Risk
0.41–0.6	Elevated Risk
0.61+	High Risk

7. Injury Types Predicted

The system initially focuses on the most common youth sports injuries.

Injury	Key Indicators
ACL injury	knee valgus + trunk instability
ankle sprain	stabilization delay
hamstring strain	asymmetry + sprint mechanics
patellar tendon injury	landing load

8. Model Training Dataset

The system will train using three datasets.

Dataset 1

AYOT screening dataset.

Collected during pilot:

- movement videos
 - screening metrics
 - injury outcomes
-

Dataset 2

Open sports biomechanics datasets.

Examples:

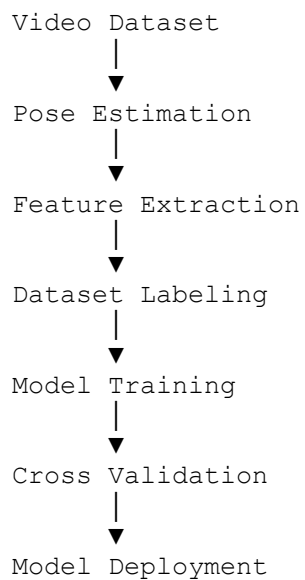
- CMU Motion Capture Dataset
 - Human3.6M
-

Dataset 3

Clinical literature thresholds.

Published biomechanical injury predictors.

9. Model Training Pipeline



10. Model Performance Metrics

Evaluate using:

Precision

Correct injury predictions.

Recall

Detected true injury cases.

ROC-AUC

Target:

ROC-AUC > 0.80

11. Clinical Validation Layer

AI outputs must always be reviewed by clinicians.

Workflow:

```
AI Risk Prediction
    ↓
Physiotherapist Review
    ↓
Final Athlete Report
```

This ensures:

- safety
 - credibility
 - regulatory compliance
-

12. Athlete Risk Intelligence Dashboard

The platform generates:

Athlete View

Risk Score
Key biomechanical issues
Corrective exercises

Team View

Team risk heatmap.

Example:

U16 Football Team

Low Risk: 8
Moderate Risk: 5
High Risk: 2

Institutional Dashboard

Displays:

- injury trends
 - athlete risk distribution
 - sports risk profiles
-

13. Continuous Learning System

Each injury recorded feeds the AI model.

Dataset structure:

```
athlete_id  
screening_metrics  
training_load  
injury_occurred  
injury_type  
time_to_injury
```

This enables **self-improving prediction accuracy**.

14. AI Infrastructure

Recommended stack.

Pose Estimation

TensorFlow MoveNet

ML Framework

PyTorch

Data Processing

Python + NumPy + Pandas

Model Serving

FastAPI

ML Pipeline

Kubeflow

Cloud

Google Cloud AI Platform

15. Long-Term AI Roadmap

Year 1

Biomechanics screening engine.

Year 2

Injury prediction models.

Year 3

Personalized athlete performance optimization.

Strategic Impact

With this AI system, AYOT SafeSport becomes:

- an **athlete injury prediction platform**
- a **sports health data company**
- a **national sports safety infrastructure provider**

Few sports medicine companies in emerging markets operate **AI-driven injury prediction systems**.

This becomes a **major competitive moat**.
